WORKED SOLUTION

STEP III 2020, Question 8

Pure •

STEP III 2020, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.8, *Counting the Rationals*: https://beyondstep.keithmcnulty.org/ch01/counting-rationals.html

The key idea

Every term with a large subscript is built from terms with about half that subscript. So any claim about the pair (u_n, u_{n+1}) can be pushed back to a pair with a smaller subscript, and an argument by *descent* (equivalently, choosing a smallest counterexample) does the work in every part. Writing out the first few terms, and how each consecutive pair produces two new ones, makes the pattern visible before any proof.

Solution

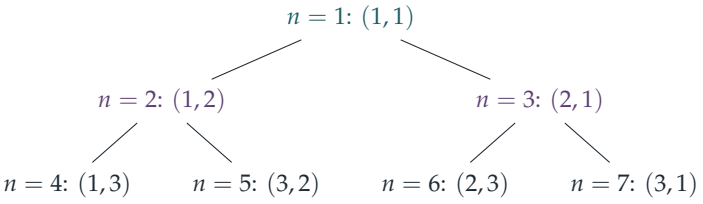
The first sixteen terms are

$$1,\ 1,\ 2,\ 1,\ 3,\ 2,\ 3,\ 1,\ 4,\ 3,\ 5,\ 2,\ 5,\ 3,\ 4,\ 1.$$

Every term is a positive integer: this holds for u_1, and each later term is either an earlier term or the sum of two earlier terms, so it follows by (strong) induction.

It is worth recording how a pair produces later pairs. If $(u_k, u_{k+1}) = (a, b)$, then directly from the definition

$$(u_{2k}, u_{2k+1}) = (a,\ a+b), \qquad (u_{2k+1}, u_{2k+2}) = (a+b,\ b). \tag{$*$}$$



The pair at position n is (u_n, u_{n+1}); its left child is at $2n$ and its right child at $2n+1$, by $(*)$. Every pair at position $n \geq 2$ arises in exactly this way from the pair at position $\lfloor n/2 \rfloor$.

Part (i)

Take a consecutive pair (u_n, u_{n+1}) with $n \geq 2$, and let $k = \lfloor n/2 \rfloor \geq 1$.

- If $n = 2k$, the pair is $(u_k,\ u_k + u_{k+1})$. The odd-subscript term $u_{2k+1} = u_k + u_{k+1}$ exceeds the even-subscript term $u_{2k} = u_k$ because $u_{k+1} \geq 1$.
- If $n = 2k+1$, the pair is $(u_k + u_{k+1},\ u_{k+1})$. Now $u_{2k+1} = u_k + u_{k+1}$ exceeds $u_{2k+2} = u_{k+1}$ because $u_k \geq 1$.

So in every pair except the first, $(u_1, u_2) = (1, 1)$, the term with odd subscript is strictly larger. In particular, *the two terms of a pair are equal only for the first pair.*

Part (ii)

Suppose $d > 1$ divides both u_n and u_{n+1}. This cannot happen for $n = 1$, since $u_1 = u_2 = 1$, so $n \geq 2$; let $k = \lfloor n/2 \rfloor$, so $1 \leq k < n$.

- If $n = 2k$: $d \mid u_{2k} = u_k$ and $d \mid u_{2k+1} = u_k + u_{k+1}$, so d divides the difference u_{k+1}.
- If $n = 2k+1$: $d \mid u_{2k+2} = u_{k+1}$ and $d \mid u_{2k+1} = u_k + u_{k+1}$, so $d \mid u_k$.

Either way d divides both u_k and u_{k+1}, an earlier consecutive pair, which is what we had to show.

Now suppose some consecutive pair has a common factor $d > 1$, and choose such a pair (u_n, u_{n+1}) with n as small as possible. As above $n \neq 1$, and the argument just given produces the pair at position $k < n$ with the same common factor, contradicting the minimality of n. Hence any two consecutive terms are co-prime.

Part (iii)

Suppose some ordered pair (a, b) occurs at two different positions, $(u_m, u_{m+1}) = (u_n, u_{n+1}) = (a, b)$ with $m < n$. Among all such repetitions, choose one with m as small as possible.

If $m = 1$ then $a = b = 1$; but $n \geq 2$, and by part (i) the pair at position n has unequal terms. So $m \geq 2$, and then also $a \neq b$ by part (i).

> **Ask yourself** How can we tell, just from a and b, whether the positions are odd or even?

Part (i) tells us that the larger term of the pair has the odd subscript, so the parity of the position is decided by the pair itself:

- If $a < b$, the second term has odd subscript, so m and n are both even: $m = 2j$,

$n = 2k$ with $j < k$. Then $u_j = u_{2j} = a$ and $u_{j+1} = u_{2j+1} - u_j = b - a$, and similarly $(u_k, u_{k+1}) = (a, b-a)$.

- If $a > b$, then m and n are both odd: $m = 2j+1$, $n = 2k+1$ with $1 \le j < k$. Then $u_{j+1} = u_{2j+2} = b$ and $u_j = u_{2j+1} - u_{j+1} = a - b$, and similarly $(u_k, u_{k+1}) = (a-b, b)$.

In both cases one ordered pair appears at the two different positions j and k, and $j \le m/2 < m$. This contradicts the choice of m. So no ordered pair of positive integers appears consecutively at two different places.

Part (iv)

Let $a > b$, with a, b co-prime, and suppose (a, b) is not a consecutive pair of the sequence. Then $a - b$ and b are positive integers, and they are co-prime: any common factor of $a - b$ and b also divides $(a - b) + b = a$, so it is a common factor of a and b and must be 1.

If $a - b$ were followed by b somewhere, say $(u_k, u_{k+1}) = (a - b, b)$, then by $(*)$

$$(u_{2k+1}, u_{2k+2}) = \big((a-b) + b,\ b\big) = (a, b),$$

contradicting the assumption on (a, b). So $(a - b, b)$ is also a co-prime pair that does not occur consecutively, and its sum is $a < a + b$.

The similar result for $a < b$: *a and $b - a$ are co-prime positive integers which do not occur consecutively with $b - a$ following a, and their sum b is smaller than $a + b$.* The proof is the same: a common factor of a and $b - a$ divides $a + (b - a) = b$, and if $(u_k, u_{k+1}) = (a, b - a)$ then $(u_{2k}, u_{2k+1}) = (a,\ a + (b - a)) = (a, b)$.

Part (v)

Since every term is a positive integer, $f(n) = u_n/u_{n+1}$ is a positive rational, so the range is contained in the positive rationals. We show that every positive rational is attained, and attained only once.

Every positive rational is in the range. Write a positive rational in lowest terms as p/q, with p, q co-prime positive integers. We claim every co-prime pair (p, q) occurs as (u_n, u_{n+1}) for some n. If not, among the co-prime pairs that do not occur choose one, (a, b), with $a + b$ as small as possible. We cannot have $a = b$: co-prime then forces $a = b = 1$, and $(1, 1)$ occurs at $n = 1$. So $a > b$ or $a < b$, and part (iv) supplies a co-prime pair that does not occur and has smaller sum. This contradicts minimality, so every co-prime pair (p, q) occurs, say at position n, and then $f(n) = p/q$.

f is one-to-one. Suppose $f(m) = f(n)$. By part (ii), u_m/u_{m+1} and u_n/u_{n+1} are both fractions in lowest terms with positive denominators. A positive rational has only one such representation, so $u_m = u_n$ and $u_{m+1} = u_{n+1}$. By part (iii) the same ordered pair cannot occur at two different places, so $m = n$.

Thus f is a bijection from the positive integers onto the positive rationals, and so it has an inverse f^{-1}: given p/q in lowest terms, $f^{-1}(p/q)$ is the unique position where p is immediately followed by q. For example, 3 is followed by 5 only at position 10, so

$f^{-1}(\frac{3}{5}) = 10$.

Checks

- The listed terms give pairs $(1,1), (1,2), (2,1), (1,3), (3,2), (2,3), (3,1), (1,4), \ldots$: odd-subscript terms are larger after the first pair, all pairs are co-prime, and no pair repeats.
- Running the descent of part (iv) on $3/5$: $(3,5) \to (3,2) \to (1,2) \to (1,1)$. Climbing back with $(*)$, $(1,1)$ at 1 gives $(1,2)$ at 2, $(3,2)$ at 5 and $(3,5)$ at 10; indeed $u_{10} = 3$, $u_{11} = 5$.
- A computer check over the first two million terms confirmed parts (i)–(iii), and found every co-prime pair (p,q) with $p+q \leq 21$ exactly once.

Common slips

- Forgetting to rule out the first pair $(1,1)$ separately in parts (ii) and (iii): it is the base case of each descent.
- In part (iii), assuming the two positions have the same parity without justifying it; part (i) is exactly what forces this.
- In part (v), proving only that the range is all positive rationals. An inverse also needs f to be one-to-one, which uses parts (ii) and (iii) together.

Beyond STEP

This is Stern's diatomic sequence, and listing u_n/u_{n+1} gives the Calkin–Wilf enumeration of the rationals: an explicit bijection between $\mathbb{N}$ and $\mathbb{Q}^+$, showing concretely that the rationals are countable. The descent in parts (iv) and (v) is the subtractive form of Euclid's algorithm, and the path from $(1,1)$ to (p,q) in the tree encodes the continued fraction of p/q. Davenport's *The Higher Arithmetic* covers Euclid's algorithm and continued fractions beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.